COMP8620

Advanced Topics in AI: Intelligent Robotics

Assignment 2 Report

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Generative AI Disclosure

OpenAI Codex was used to prepare the report template, review and revise the Part A answers, implement the first Part B solver and tests, design and run the first experiments, and draft report text, pseudocode and figures. Anthropic Claude Code (Claude Opus 5.5) was then used to review the Part B code and experiments, implement the revised solver (risk-aware guidance and admissible exploration), redesign and rerun the Part B experiments, analyse the measured results, and revise the Part B text, tables and figures. The supplied simulator's transition and reward methods were preserved; their use and the headless import changes are documented in Part B. Original-language prompts from all these interactions, including supplied draft answers, are recorded verbatim in `AI_prompts.txt` in the submitted source archive.

Part A: Theory (30 points)

Question 1: Two-state MDP abstraction (10 points)

No.

Let $m = \lceil k/2 \rceil$, and define

$$\phi(s) = \begin{cases} L, & 1 \le s \le m, \\ R, & m+1 \le s \le k. \end{cases}$$

An exact MDP abstraction requires a transition function depending only on the abstract state and action while preserving the original motion probabilities. In particular, for every action a, transition consistency requires

$$\phi(s_1) = \phi(s_2) \implies \sum_{u:\, \phi(u)=R} T(s_1, a, u) = \sum_{u:\, \phi(u)=R} T(s_2, a, u).$$

For $k > 4$, consider $s_1 = m - 1$ and $s_2 = m$, both in L. Under $a = \text{moveRight}$,

$$m-1 \xrightarrow{0.7} m \in L, \qquad m-1 \xrightarrow{0.3} m-2 \in L,$$
$$m \xrightarrow{0.7} m+1 \in R, \qquad m \xrightarrow{0.3} m-1 \in L.$$

Hence,

$$P(R \mid s_1, a) = \sum_{u=m+1}^{k} T(m-1, a, u) = 0,$$
$$P(R \mid s_2, a) = \sum_{u=m+1}^{k} T(m, a, u) = T(m, a, m+1) = 0.7.$$

Since $0 \neq 0.7$, transition consistency fails: no single $T'(L, \text{moveRight}, R)$ can represent both states exactly.

Therefore, the proposed abstraction cannot preserve the original motion uncertainty and is not an exact MDP abstraction.

Question 2: Alpha-vector dominance (10 points)

Let $p = b(s_1)$. Since $b(s_1) + b(s_2) = 1$,

$$b = (p, 1-p), \qquad 0 \le p \le 1.$$

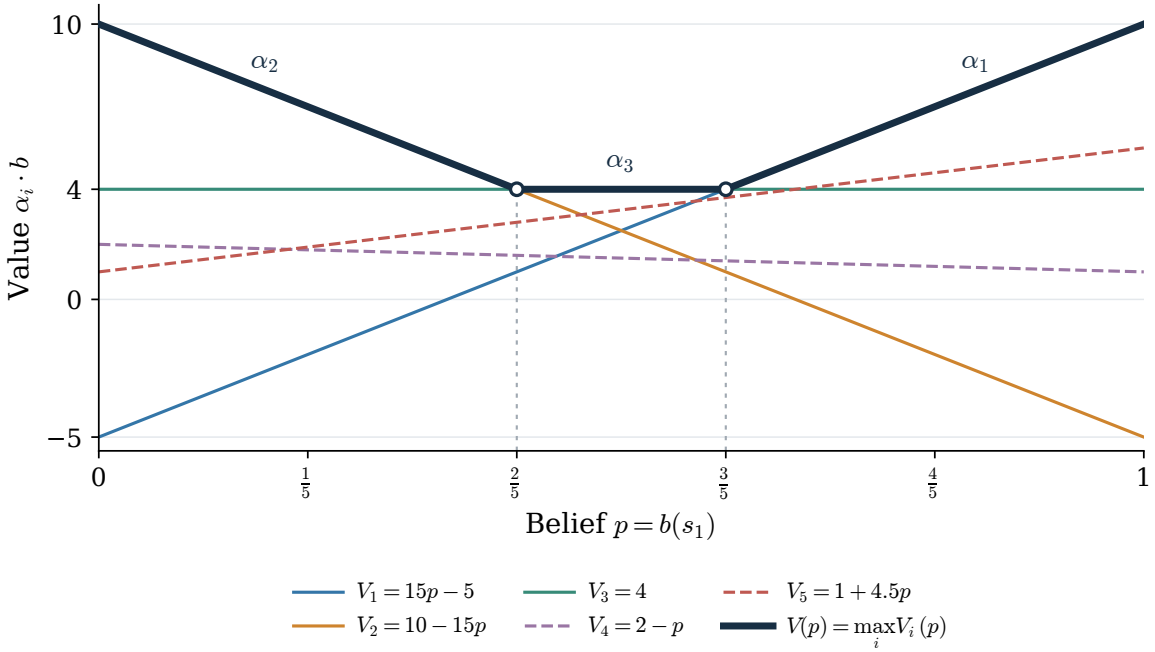
For each $\alpha_i \in \Gamma$, let $V_i(p) = \alpha_i \cdot b$. Then

$$\begin{aligned} V_1(p) &= 10p - 5(1-p) = 15p - 5, \\ V_2(p) &= -5p + 10(1-p) = 10 - 15p, \\ V_3(p) &= 4p + 4(1-p) = 4, \\ V_4(p) &= p + 2(1-p) = 2 - p, \\ V_5(p) &= 5.5p + (1-p) = 1 + 4.5p. \end{aligned}$$

(a) Maximal belief intervals (4 points)

The upper envelope is

$$V(p) = \max_{\alpha_i \in \Gamma} V_i(p).$$



The relevant intersections are

$$\begin{aligned} V_2(p) = V_3(p) &\iff 10 - 15p = 4 \iff p = \frac{2}{5}, \\ V_1(p) = V_3(p) &\iff 15p - 5 = 4 \iff p = \frac{3}{5}. \end{aligned}$$

The remaining vectors are never maximal, since

$$V_4(p) = 2 - p < 4 = V_3(p), \qquad 0 \le p \le 1,$$

and, using $V_1(p) - V_5(p) = 10.5p - 6$,

$$\begin{cases} V_5(p) \le 1 + 4.5 \times \frac{3}{5} = 3.7 < 4 = V_3(p), & 0 \le p \le \frac{3}{5}, \\ V_1(p) - V_5(p) \ge 10.5 \times \frac{3}{5} - 6 = 0.3 > 0, & \frac{3}{5} \le p \le 1. \end{cases}$$

Since V_2 decreases, V_1 increases, and $V_3 = 4$,

$$V(p) = \begin{cases} 10 - 15p, & 0 \le p < \frac{2}{5}, \\ 4, & \frac{2}{5} \le p \le \frac{3}{5}, \\ 15p - 5, & \frac{3}{5} < p \le 1. \end{cases}$$

Therefore, the maximal belief intervals are

$$\boxed{\begin{aligned} \alpha_1 &: \left[\frac{3}{5}, 1\right], \\ \alpha_2 &: \left[0, \frac{2}{5}\right], \\ \alpha_3 &: \left[\frac{2}{5}, \frac{3}{5}\right], \\ \alpha_4, \alpha_5 &: \varnothing. \end{aligned}}$$

At $p = \frac{2}{5}$, α_2 and α_3 tie; at $p = \frac{3}{5}$, α_1 and α_3 tie.

(b) Pointwise dominance (6 points)

An α-vector can be removed if, at every belief, another vector has strictly greater value:

$$\forall b \in B, \quad \exists \alpha' \in \Gamma : \alpha \cdot b < \alpha' \cdot b.$$

For α_4,

$$V_4(p) = 2 - p < 4 = V_3(p), \qquad \forall p \in [0, 1].$$

Thus, α_4 is strictly dominated by α_3.

For α_5, the dominating vector may depend on p. From part (a),

$$\begin{cases} V_5(p) < V_3(p), & 0 \le p \le \frac{3}{5}, \\ V_5(p) < V_1(p), & \frac{3}{5} < p \le 1. \end{cases}$$

These intervals cover $[0, 1]$, so α_5 also satisfies the removal condition.

The remaining vectors α_2, α_3, and α_1 are uniquely maximal at $p = 0$, $p = \frac{1}{2}$, and $p = 1$, respectively. Hence, none satisfies the removal condition.

Therefore,

$$\boxed{\Gamma_{\text{remove}} = \{\alpha_4, \alpha_5\}, \qquad \Gamma_{\text{keep}} = \{\alpha_1, \alpha_2, \alpha_3\}.}$$

Question 3: MDP versus POMDP for pedestrian avoidance (10 points)

Let c_t, v_t be the car's position and speed, and p_t^i the position of pedestrian i. Define

$$x_t = (c_t, v_t, p_t^1, \ldots, p_t^n).$$

Positions are observed accurately, and the car measures its own speed. It chooses an action a_t (accelerate, maintain speed, or brake) to reach G while avoiding collisions.

(a) Scenario suitable for an MDP

Suppose pedestrians follow known stochastic movement rules: the probabilities of crossing or continuing along the sidewalk depend only on the current observable configuration, with no persistent hidden intention. These memoryless dynamics give

$$P(x_{t+1} \mid x_{0:t}, a_{0:t}) = P(x_{t+1} \mid x_t, a_t).$$

Thus, x_t is Markov and fully observed, so an MDP with state $s_t = x_t$ is suitable despite the stochastic pedestrian motion.

(b) Scenario suitable for a POMDP

Now suppose each pedestrian has a persistent but unobservable destination g_i, such as crossing the road or continuing along the sidewalk. Let $\mathbf{g} = (g_1, \ldots, g_n)$, and assume motion is Markov conditional on these destinations. The complete state and observation are

$$s_t = (x_t, \mathbf{g}), \qquad o_t = x_t.$$

At the same observed configuration, two possible joint destinations can yield different motion distributions:

$$P(x_{t+1} \mid x_t, a_t, \mathbf{g}^{(1)}) \neq P(x_{t+1} \mid x_t, a_t, \mathbf{g}^{(2)}).$$

Different motion histories can produce the same x_t but different beliefs about $\mathbf{g}$. Hence, x_t alone need not be Markov. The car maintains

$$b_t(\mathbf{g}) = P(\mathbf{g} \mid x_{0:t}, a_{0:t-1})$$

and predicts motion using

$$P(x_{t+1} \mid x_t, b_t, a_t) = \sum_{\mathbf{g}} P(x_{t+1} \mid x_t, a_t, \mathbf{g})\, b_t(\mathbf{g}).$$

The belief is updated as new movements are observed. A POMDP is suitable because it uses both the observed configuration and the belief over hidden destinations to select driving actions.

Part B: Multi-drone planning under uncertainty (35 points)

Question 1: MDP formulation (3 points)

Let $\mathcal{G} = \{0, \dots, n_x - 1\} \times \{0, \dots, n_y - 1\} \times \{0, \dots, n_z - 1\}$, with obstacles O and known drone goals g_i. The finite-horizon MDP is $(\mathcal{S}, \mathcal{A}, T, R, \gamma)$.

States and actions. A live state is

$$s = ((p_i, r_i)_{i=1}^{N}, t), \qquad 0 \le t < H,$$

where p_i is drone i's cell and r_i indicates that it has reached its goal. Reached drones remain at their goals and occupy those cells. The state space also contains an absorbing state $\perp$, entered after a collision, all goals being reached, or the step limit H. Each active drone commands one of $B = 26$ neighbour offsets (or $B = 8$ with fixed altitude), giving $\mathcal{A} = \{0, \dots, B-1\}^N$. There is no hover action; reached drones ignore their commands.

Transitions. Let $q_i(y_i \mid p_i, r_i, a_i)$ be the supplied, normalised single-drone target law (Appendix A.1). For a reached drone, $q_i(y_i \mid p_i, 1, a_i) = \mathbf{1}\{y_i = p_i\}$. Raw targets $y = (y_1, \dots, y_N)$ are conditionally independent. Let $F(s, y)$ update positions and flags, increment t, and return $\perp$ on termination. Then

$$T(s, a, s') = \sum_{y : F(s,y) = s'} \prod_{i=1}^{N} q_i(y_i \mid p_i, r_i, a_i), \qquad T(\perp, a, \perp) = 1.$$

An obstacle hit or shared returned cell terminates the mission; cell swaps and diagonal crossings are permitted by the supplied collision rule.

Reward and objective. For a sampled outcome y, let K_O count obstacle hits, K_V count drones in shared-cell groups, and Δr_i indicate a newly reached goal. The simulator returns

$$\widetilde{r}(s, a, y) = N c_{\text{step}} + C(K_O + K_V) + G \sum_i \Delta r_i, \qquad c_{\text{step}} < 0,\ C < 0,\ G > 0.$$

Thus $R(s, a) = \mathbb{E}_y[\widetilde{r}(s, a, y) \mid s, a]$ and $R(\perp, a) = 0$. The terminal transition receives its reward before absorption. The objective is

$$\max_{\mu}\ \mathbb{E}_{\mu}\left[\sum_{t=0}^{\tau-1} \gamma^t \widetilde{r}_t\right], \qquad \tau = \min\{T_{\text{success}}, T_{\text{collision}}, H\}.$$

This encourages fast completion and penalises collisions.

Question 2: Approximate online solver (4 points)

I use UCT/MCTS with progressive widening, re-planned after every executed step. The supplied generative model samples outcomes without enumerating the state space or all $26^4 = 456{,}976$ joint actions.

Guidance. Cached Dijkstra maps J_i trade path length against one-step obstacle-slip risk. For a command from p_i towards q, the score is

$$J_i(q) + 0.025\|q - g_i\|_1 + \frac{0.3}{\max\{1, d(q)\}} + \lambda \ell(p_i) \sum_{\substack{k \neq i \\ r_k = 0}} \sigma(p_k) \mathbf{1}\{q \in \mathcal{N}(p_k)\}.$$

Here d is the supplied obstacle-distance transform; ℓ and σ are the normalised intended and per-neighbour slip probabilities, and λ converts collision loss into step-cost units (Appendix A.1).

The last term is an inter-drone risk surrogate. $\mathcal{N}$ contains physical neighbours at the allowed altitude. Out-of-grid and obstacle targets receive prohibitive costs; frozen or already reserved targets incur an additional 10^9. GREEDY processes drones in index order and minimises this score; SAMPLE uses random order and samples among the four cheapest commands within 1.05 of the minimum, with weights $e^{-3(\text{score}-\min)}$. EXPLORE samples uniformly from free, in-grid, unreserved targets when available.

The four MCTS components.

(1) **Selection.** Try unvisited admitted actions first; otherwise maximise

$$Q(s,a) + c_{\text{UCT}}\sqrt{\frac{\log(N(s)+1)}{N(s,a)}}, \qquad c_{\text{UCT}} = 0.35.$$

Returns are normalised by $\kappa = \max\{1, N(G + 2|C|)\}$.

(2) **Expansion.** The action limit is $\min\{\lceil 0.5\sqrt{N(s)+1}\rceil, \prod_{i:r_i=0} |U_i^{\text{adm}}|\}$. Here U_i^{adm} contains commands towards free, in-grid neighbours. Admit GREEDY first, then EXPLORE with probability 0.15 and SAMPLE otherwise (at most 12 attempts for a new action). Each distinct sampled successor has its own child.

(3) **Simulation.** Roll out for at most eight steps, choosing EXPLORE with probability 0.1, otherwise GREEDY. Terminal outcomes have zero continuation value; other cutoffs add

$$\widehat{V}(s,h) = Nc_{\text{step}} \sum_{j=0}^{D-1} \gamma^j + G \sum_{\substack{i:r_i=0 \\ 1\le\delta_i\le h}} \gamma^{\delta_i - 1}, \qquad D = \min\{h, \max_{i:r_i=0} \delta_i\}.$$

δ_i is the physical breadth-first distance, set to ∞ when unreachable or when frozen goal occupants disconnect the route. The latter connectivity check is cached on maps of at most 20,000 cells. Tree depth is capped at eight.

(4) **Backpropagation.** Back up $v = R_{\text{sample}}/\kappa + \gamma v'$; increment edge and node visits and update the edge mean and sample variance using Welford's recurrence.

Root decision. Let a_g be the greedy action and $\hat{\sigma}^2$ the pooled within-action variance. Among visited actions, define

$$m(a) = Q(a) - Q(a_g) - z_{\text{sw}}\hat{\sigma}\sqrt{\frac{1}{N(a)} + \frac{1}{N(a_g)}}, \qquad z_{\text{sw}} = 1.$$

Execute the action with largest positive $m(a)$; otherwise keep a_g. With no completed trial, use a_g directly. This variance-aware switching heuristic has no formal policy-improvement guarantee. Parameters were selected on 16 separate development layouts; the tuning results are archived in `results/dev/`.

Listing 1: Online UCT; rollout and admission rules are defined above.

```
PROCEDURE PLAN(s, budget, K = 2000)
    deadline <- NOW + max(0, 0.97 * min(budget, 10) - 0.002)
    a_g <- GREEDY(s); root <- empty node; h <- H - t; sims <- 0
    WHILE NOW < deadline AND sims < K AND h > 0 AND not all reached:
        TRY: SEARCH(root, s, h, 0); sims += 1
        ON Deadline: BREAK
    RETURN root decision above, or a_g if no trial completed

PROCEDURE SEARCH(node, s, h, depth)
    IF h = 0 OR all reached: RETURN 0
    IF depth >= 8: RETURN ROLLOUT(s, h)
    admit an action under the expansion rule, if permitted
    a <- unvisited action first, otherwise argmax UCB
    (s', r, done) <- SIMULATE(s, a) // check deadline before call
    IF done: v <- 0
    ELSE IF child(node, a, s') is new:
        create child; v <- ROLLOUT(s', h - 1)
    ELSE: v <- SEARCH(child(node, a, s'), s', h - 1, depth + 1)
    v <- r / kappa + gamma * v
    update visits, mean and variance; RETURN v
```

ROLLOUT discounts sampled rewards and the terminal/cutoff value defined above. It may stop at the deadline and return a leaf estimate; a deadline exception before a simulator call aborts that trial. The planner tracks the remaining horizon because `simulate` does not enforce the execution step limit.

Question 3: Implementation and demonstration (8 points)

`run_u8076655.py` implements B2 using the supplied MultiDroneUnc simulator's unchanged dynamics, with independent planning and execution RNGs. From the extracted archive's top-level folder:

Reproduction commands.

```
# Dependencies (NumPy, SciPy, PyYAML; Matplotlib only for figures).
python -m pip install -r requirements.txt

# Complete template; arguments in brackets are optional.
python run_u8076655.py --config <scenario.yaml> \
  [--planning-time SECONDS] [--max-simulations K] [--seed SEED] \
  [--visualize] [--quiet] [--output TRACE.json] \
  [--variant VARIANT] [--action-mask MASK]
#   --planning-time    seconds per decision (default 9.5, max 10)
#   --max-simulations  MCTS trials per decision (default 2000)
#   --seed             execution seed (default: YAML seed, else 0)
#   --visualize        Vedo 3D viewer (headless if vedo is missing)
#   --quiet            no per-step progress lines
#   --output           save trajectory and search diagnostics (JSON)
#   --variant          B4 ablation (default full = submitted solver)
#   --action-mask      B4 command restriction (default full)

# Minimal template (the marking command).
python run_u8076655.py --config <scenario.yaml>

# Example.
python run_u8076655.py --config environments/demo_four_3d.yaml --visualize
```

Each executed step prints one line to standard error; a JSON summary goes to standard output. Excerpt from the open 2D validation run below:

```
step   1/80 | 0/1 at goal | reward    -1.0 | plan  9.14s  2000 sims | (1,3,0)
step   2/80 | 0/1 at goal | reward    -1.0 | plan  8.68s  2000 sims | (2,4,0)
...
step  11/80 | 1/1 at goal | reward    99.0 | plan  1.87s  2000 sims | (11,5,0)*
success: True (success), steps: 11, total discounted reward: 71.74
```

Planning is serial, with a 9.5 s allowance and 2,000-trial cap; the deadline is checked before each simulator call. All 35 unit tests pass. Local single-process runs with the default 9.5-second allowance (seed 0) give:

Task	Status	Steps	Sims/step	Mean (s)	Max (s)
Open 2D, 1 drone, 8 actions	success	11	2,000	5.37	9.14
Supplied example, 2 drones	success	17	2,000	5.43	7.39
Strong wind, 1 drone, $\alpha = 3$	success	18	2,000	4.69	6.29
Four 3D, 5 obstacles	success	13	1,983	6.34	9.21

Question 4: Experimental investigation (20 points)

Experimental design

The main evaluation uses 2,000 simulations per decision; S3 includes ablations and a zero-search policy, and S4 varies the simulation cap. Experimental runs use a relaxed wall-clock cap to isolate search effort. The demo instead stops at the first of its time and trial limits, so these are different budget protocols. Conditions share execution seeds 0–19 unless noted, with comparisons paired within each scenario. Development layouts were excluded from evaluation. Only independent experimental episodes ran in parallel; planning within each episode was serial.

Set	Controlled scenarios	Question and expectation
S1 (540)	3^3 factorial: $12 \times n_y \times 3$ grids, $n_y = 5, 10, 20$; 6/18/26 commands per drone ($\|A\| = 36/324/676$); 0/2/5 nested obstacles. Two drones, $\alpha = 0.25$, $H = 60$.	Rank the three factors and test interactions. Off-route cells should have little effect; richer control may help; clutter adds detours and slip risk.
S2 (160)	S1 clutter layouts at $\alpha = 0.01$ with 6/26 commands; duplicate command labels $\times 2, \times 4$ without changing physics.	Separate geometric detours from noise, and control authority from branching.
S3 (880)	Five ablations and the search-free policy at all eight low/high S1 corners. Uniform proposals use ten seeds; others use twenty.	Measure how component effects change with each complexity factor.
S4 (400)	16/64/256/1,024/2,000 simulations on open 1-drone, strong-wind 1-drone, cluttered 2-drone and 3D 4-drone tasks.	Test whether additional search improves completion as tasks get harder.
S5 (160)	$N = 1, 2, 3, 4$ crossed with $\alpha = 0.05, 0.25, 1, 2$; one random $8 \times 6 \times 3$ layout per pair, ten seeds each.	Check performance beyond the factorial and joint growth of state/action spaces.

Table 1: Evaluation sets; parentheses give episode counts.

A further 360 regression, 520 search-free baseline and 280 supplementary runs provide additional validation (Appendix A.5); the complete evaluation contains 3,300 episodes.

Metrics. With F free cells, there are up to F^N positional configurations and B^N joint commands. Success means all goals reached within H without collision. Penalised completion length is $L =$ steps on success and $L = H$ on failure. Success proportions have Wilson 95% intervals; continuous means have Student-t intervals $\bar{x} \pm t_{0.975,n-1} s/\sqrt{n}$. High-minus-low factor effects average the other factors within each seed; component effects pair each ablation with the full solver before averaging corners. Their 95% intervals use seed blocks. Supporting means and intervals are in Appendix A.

S1: effects of the three complexity factors

Factor (high minus low)	ΔL (95% CI)	Δsuccess (pp; 95% CI)
State space (n_y 5→20)	+0.0 [-1.4, +1.4]	+1.7 [-1.4, +4.7]
Action space ($\|A\|$ 36→676)	-5.5 [-7.0, -4.1]	+0.6 [-2.6, +3.7]
Clutter ($\|O\|$ 0→5)	+7.1 [+5.7, +8.6]	-5.0 [-8.1, -1.9]

Table 2: Paired factor effects; positive ΔL indicates degradation.

Clutter causes the largest degradation within these tested levels. It is the only factor whose success-effect interval excludes zero. The solver succeeds in 97.6% of S1 episodes; all 13

failures are collisions and none reaches the horizon.

State-space padding has little observed effect. For $n_y = 10$ versus 20, 120 of 180 pairs match in steps, return and outcome. This tests off-route padding, not longer routes or additional drones.

Richer commands shorten routes. Changing 6 to 26 directions reduces L; the success difference is unresolved. The action–clutter interaction is -7.7 $[-11.0, -4.4]$ steps, consistent with diagonal/vertical moves easing obstacle avoidance. Neither state-related interaction is resolved ($S \times C$: $[-3.8, +4.7]$; $S \times A$: $[-3.1, +4.3]$).

S2: geometry, noise and action branching

At $n_y = 10$, the clutter effect averaged over the two command sets is +9.4 steps $[+6.4, +12.5]$ at $\alpha = 0.25$, versus +1.5 $[+1.2, +1.8]$ at $\alpha = 0.01$, where all episodes succeed. Thus geometry still adds detour cost, while noise substantially increases the observed clutter penalty. The supplied transition law raises slip probability near obstacles.

Duplicating command labels up to $|A| = 10{,}816$ changes L by only +0.1 $[-0.1, +0.2]$ and leaves the admitted root-action count near 23. This separates pure label branching from the extra control authority in S1. Detailed controls, including the visit share of the *executed* action, appear in Appendix A.3.

S3: component effects as complexity increases

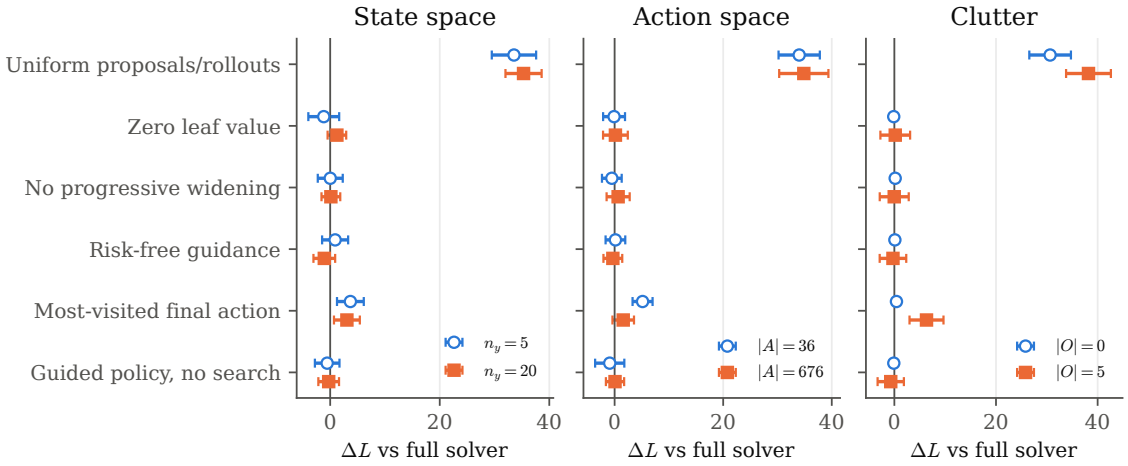


Figure 1: Paired change in L after removing a component at low/high factor levels (open/filled markers); 95% seed-block intervals. Positive values favour the full solver.

Guidance in expansion and rollout has the largest measured effect. Uniform proposals/rollouts raise L by +34.4 $[+31.5, +37.4]$ and change success by -50.0 pp. The loss grows from +30.6 without obstacles to +38.2 with five; the clutter interaction excludes zero ($[+1.4, +13.7]$), whereas its state/action interactions do not. This ablation removes both route guidance and admissible-target filtering.

Root choice also matters: most-visited selection raises L by +3.4 $[+1.7, +5.0]$. The increase is larger with clutter ($[+2.5, +9.3]$ interaction interval) and smaller with richer commands ($[-5.5, -1.6]$). These comparisons favour the chosen rule on these tasks.

Widening, leaf estimates and risk surcharges have no statistically resolved effects across these corners (full estimates in Appendix A.4). Routes are roughly eleven steps, comparable to

the sampled tree depth plus eight-step rollout, which may limit the leaf estimate's contribution. The risk-free ablation retains route guidance; its result therefore does not contradict the loss from removing all guidance.

Search versus its base policy. Removing search changes L by -0.4 $[-1.7, +0.9]$, with no resolved interaction. The evaluated completion performance is largely explained by the guided base policy. In the additional regression/generated comparison, multi-drone search rescues 12 episodes and loses 4; the pooled difference is inconclusive (Appendix A.5).

S4: search budget

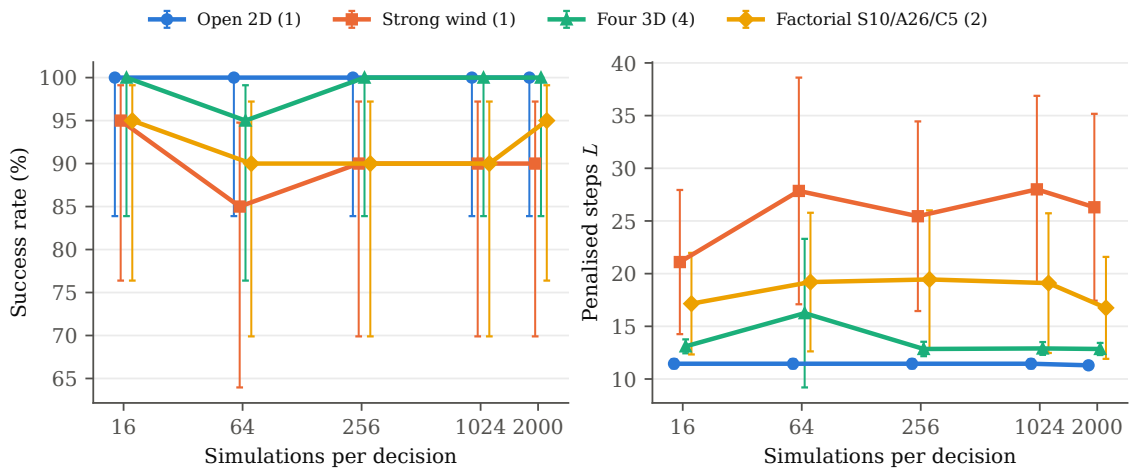


Figure 2: Success and L versus simulations per decision; 20 runs per point, with Wilson and t 95% intervals. Drone counts are in parentheses.

There is no consistent improvement from 16 to 2,000 simulations. For four drones, success is 20/20 and 20/20 respectively, while mean new-node depth increases from 1.9 to 2.7. More search changes the tree without a demonstrated completion benefit here; overlapping intervals do not establish budget equivalence.

S5: generated layouts

Across the crossed drone-count/noise set, 148/160 episodes succeed. At $\alpha \leq 0.25$, all 80/80 runs succeed, including 20/20 with four drones; four drones at $\alpha = 2$ succeed in 3/10. This indicates sensitivity to noise in the tested layouts, while increasing N jointly enlarges state and action spaces. Mean new-node depth is 3.1 for one drone and 2.4 for four, consistent with less successor reuse when joint outcomes multiply. Each (N, α) pair has only one layout, so the per-layout results in Appendix A.6 do not isolate layout effects.

Limitations

The factorial covers two drones at $\alpha = 0.25$, and S5 has only one layout per (N, α) pair. The t intervals for bimodal L are approximate. Fixed-simulation runs do not reproduce a binding demo deadline. Static guidance approximates detours around frozen drones. Successful wall-gap runs average 39 steps for an 11-step static route, suggesting an optimistic cutoff estimate; these experiments do not isolate the cause.

A Supporting definitions and results

Complete regression, budget and tuning data remain in the source archive. The six original evaluation suites retain their generating hashes; all 137 original seed-0 traces replay identically with the current planner (`results/planner_equivalence.json`). Baseline and supplementary runs use the current planner directly.

A.1 Algorithm definitions

For a free in-grid target command, let $\mathcal{N}(q)$ be the physical neighbours of q and $V(q) = |\mathcal{N}(q)|$. The supplied motion law gives

$$\pi(q) = \frac{d(q)}{d(q)+\alpha}, \quad M(q) = \pi(q) + (V(q)-1)\frac{1-\pi(q)}{B-1},$$
$$\ell(q) = \frac{\pi(q)}{M(q)}, \qquad \sigma(q) = \frac{1-\pi(q)}{(B-1)M(q)}.$$

Dijkstra entry costs are $1 + \lambda\rho(q)$ for a non-goal cell and 1 for the goal, where

$$\rho(q) = |\mathcal{N}(q) \cap O|\sigma(q), \qquad \lambda = \frac{w(G+|C|)}{\max\{|c_{\text{step}}|, 10^{-3}(G+|C|), 10^{-9}\}}, \quad w = 1.$$

With $w = 0$, these are breadth-first distances. The supplied empty-map distance transform uses a virtual point beyond the grid corner, so slips remain possible without obstacles.

A rollout cut off after L sampled steps returns

$$\sum_{j=0}^{L-1} \gamma^j r_j/\kappa + \gamma^L \widehat{V}(s_L, h-L)/\kappa,$$

with zero continuation on termination. For an edge sample v, update

$$n \leftarrow n+1, \quad \Delta \leftarrow v - Q, \quad Q \leftarrow Q + \Delta/n, \quad S \leftarrow S + \Delta(v-Q).$$

The pooled root variance is $\hat{\sigma}^2 = \sum_a S_a / \sum_a (n_a - 1)$ over visited actions, or zero when the denominator is zero.

A.2 Factorial results

Table 3: All 27 S1 conditions, $n = 20$ each. L and Return: mean ±95% interval half-width; Calls: mean simulator calls per decision.

n_y	$\lvert A\rvert$	$\lvert O\rvert$	Success	L	Return	Calls
5	36	0	20/20	13.3 ± 0.9	134.1 ± 3.9	18,413
5	36	2	19/20	21.8 ± 4.6	99.0 ± 23.0	19,010
5	36	5	18/20	23.4 ± 6.1	89.1 ± 30.9	18,539
5	324	0	20/20	12.1 ± 0.6	140.0 ± 2.1	14,951
5	324	2	19/20	15.8 ± 4.9	117.8 ± 35.1	16,166
5	324	5	18/20	18.1 ± 6.7	110.7 ± 34.1	15,664
5	676	0	20/20	12.1 ± 0.7	140.0 ± 2.2	14,730
5	676	2	20/20	13.2 ± 0.9	135.7 ± 3.1	15,330
5	676	5	19/20	16.2 ± 4.9	120.7 ± 24.7	15,284
10	36	0	20/20	13.8 ± 0.7	133.0 ± 2.5	18,368
10	36	2	20/20	17.9 ± 1.2	116.0 ± 4.6	18,599
10	36	5	19/20	28.1 ± 3.9	75.5 ± 28.1	19,998
10	324	0	20/20	12.1 ± 0.6	139.8 ± 2.0	14,939
10	324	2	20/20	13.2 ± 0.7	135.7 ± 2.5	15,738
10	324	5	20/20	15.7 ± 0.8	125.4 ± 3.1	17,113
10	676	0	20/20	12.2 ± 0.6	139.5 ± 2.1	14,759
10	676	2	20/20	13.0 ± 0.7	136.7 ± 2.3	15,264
10	676	5	19/20	16.8 ± 4.8	115.2 ± 33.6	16,457
20	36	0	20/20	13.8 ± 0.7	133.0 ± 2.5	18,368
20	36	2	19/20	20.0 ± 4.6	104.8 ± 23.2	18,891
20	36	5	20/20	25.4 ± 1.4	90.0 ± 3.5	19,787
20	324	0	20/20	12.1 ± 0.6	139.8 ± 2.0	14,939
20	324	2	20/20	13.1 ± 0.7	135.6 ± 2.9	15,738
20	324	5	19/20	17.8 ± 4.7	114.5 ± 24.0	17,184
20	676	0	20/20	12.2 ± 0.6	139.5 ± 2.1	14,759
20	676	2	19/20	15.4 ± 5.0	119.6 ± 35.3	15,672
20	676	5	19/20	16.4 ± 4.9	120.1 ± 24.6	16,513

A.3 Mechanism controls

Command masks restrict intended directions; the physical slip law retains all 26 neighbours. Duplicated labels leave the physical commands unchanged.

Table 4: S2 clutter controls ($n_y = 10$, $n = 20$). L: mean ±95% interval half-width; Obst./Veh.: collision episodes.

Commands	α	$\lvert O\rvert$	Success	L	Obst.	Veh.
all 26	0.25	0	20/20	12.2 ± 0.6	0	0
		2	20/20	13.0 ± 0.7	0	0
		5	19/20	16.8 ± 4.8	0	1
	0.01	0	20/20	11.0 ± 0.0	0	0
		2	20/20	11.0 ± 0.0	0	0
		5	20/20	11.2 ± 0.2	0	0
axial 6	0.25	0	20/20	13.8 ± 0.7	0	0
		2	20/20	17.9 ± 1.2	0	0
		5	19/20	28.1 ± 3.9	0	1
	0.01	0	20/20	11.6 ± 0.4	0	0
		2	20/20	13.2 ± 0.3	0	0
		5	20/20	14.4 ± 0.5	0	0

Table 5: Duplicated labels ($n_y = 10$, $|O| = 2$, $n = 20$). Root: admitted actions; Selected: visit share of the executed action. L: mean ±95% interval half-width.

Labels	$\|A\|$	Success	L	Root	Selected (%)
×1	676	20/20	13.0 ± 0.7	22.6	5.4
×2	2,704	20/20	13.1 ± 0.9	23.0	5.2
×4	10,816	20/20	13.1 ± 0.8	23.0	5.1

A.4 Component effects

Table 6: Paired S3 effects. Overall changes include 95% intervals; factor columns give low/high-level means. *: interaction interval excludes zero. Uniform proposals: ten seeds; others: twenty.

Variant	ΔL	Δsuccess (pp)	n_y 5 / 20	$\|A\|$ 36 / 676	$\|O\|$ 0 / 5
Uniform proposals/rollouts	+34.4 [+31.5, +37.4]	-50.0 [-56.0, -44.0]	+33.5 / +35.3	+34.0 / +34.9	+30.6 / +38.2*
Zero leaf value	+0.0 [-1.5, +1.5]	+0.6 [-2.4, +3.6]	-1.2 / +1.2	-0.1 / +0.1	-0.1 / +0.2
No progressive widening	+0.1 [-1.4, +1.5]	+0.0 [-3.3, +3.3]	+0.0 / +0.1	-0.5 / +0.7	+0.2 / -0.0
Risk-free guidance	-0.1 [-1.4, +1.2]	-1.9 [-4.7, +1.0]	+0.9 / -1.1	+0.1 / -0.3	+0.1 / -0.3
Most-visited final action	+3.4 [+1.7, +5.0]	-1.9 [-6.2, +2.5]	+3.7 / +3.0	+5.2 / +1.6*	+0.4 / +6.3*
Guided policy, no search	-0.4 [-1.7, +0.9]	-0.6 [-3.6, +2.4]	-0.6 / -0.3	-0.9 / +0.1	-0.1 / -0.7

A.5 Additional validation

Among 240 feasible multi-drone regression/generated pairs, search succeeds in 226 versus 218; removing search changes L by $+1.3$ $[-0.4, +3.0]$. These exploratory intervals pool episode differences across layouts. The supplementary cases below include fixed-altitude single-drone maps, separate altitude layers and a goal inside a teammate's corridor. Wall-gap and corridor cases use seven/eight obstacles, beyond the demo's five-obstacle range. No timeout occurred; all failures were collisions.

Table 7: Supplementary cases ($n = 20$ per policy). No search: paired base policy; Obst./Veh./T.o.: full-solver failure counts. L: mean ±95% interval half-width.

Case	Success	Wilson 95% (%)	No search	Obst.	Veh.	T.o.	L
Supplied map, 1 drone	20/20	[84, 100]	20/20	0	0	0	11.2 ± 0.8
Wall with one gap, 1 drone	13/20	[43, 82]	14/20	7	0	0	53.1 ± 11.7
Slalom, 1 drone	19/20	[76, 99]	18/20	1	0	0	28.6 ± 6.7
Five obstacles, 1 drone	20/20	[84, 100]	15/20	0	0	0	16.8 ± 1.8
Two layers, crossing, 2 drones	20/20	[84, 100]	19/20	0	0	0	13.6 ± 1.5
Two layers, 4 drones	20/20	[84, 100]	20/20	0	0	0	12.5 ± 0.9
Goal inside corridor, 2 drones	13/20	[43, 82]	10/20	7	0	0	36.9 ± 15.3

A.6 Generated-layout results

Table 8: All 16 S5 layouts, one per (N, α) pair, ten seeds each. L: mean ±95% interval half-width; Depth: mean new-node depth.

N	α	Success	Wilson 95% (%)	L	Depth
1	0.05	10/10	[72, 100]	7.1 ± 0.2	3.80
1	0.25	10/10	[72, 100]	7.3 ± 0.5	3.45
1	1	9/10	[60, 98]	14.2 ± 11.6	2.78
1	2	10/10	[72, 100]	12.9 ± 2.4	2.32
2	0.05	10/10	[72, 100]	7.6 ± 0.6	3.48
2	0.25	10/10	[72, 100]	8.1 ± 1.1	3.11
2	1	9/10	[60, 98]	16.4 ± 11.0	2.27
2	2	9/10	[60, 98]	23.0 ± 10.7	2.01
3	0.05	10/10	[72, 100]	7.9 ± 0.6	3.28
3	0.25	10/10	[72, 100]	10.2 ± 1.2	2.60
3	1	10/10	[72, 100]	11.1 ± 1.1	2.03
3	2	10/10	[72, 100]	17.5 ± 1.6	1.71
4	0.05	10/10	[72, 100]	7.5 ± 0.6	3.23
4	0.25	10/10	[72, 100]	8.2 ± 0.7	2.94
4	1	8/10	[49, 94]	23.6 ± 13.8	1.85
4	2	3/10	[11, 60]	46.5 ± 15.6	1.41

B Example trajectories and planning times

Stored seed-0 full-solver traces illustrate scenario types, including a collision; they do not replace B4's repeated-run statistics. Squares mark starts, stars goals, and crosses unfinished drones' returned positions. Collision targets are not logged. Only the first sheet uses the demo's 9.5 s allowance; later sheets use fixed-simulation experiments with relaxed deadlines. The complete gallery covers all 76 evaluation configurations and the supplied example, excluding development layouts.

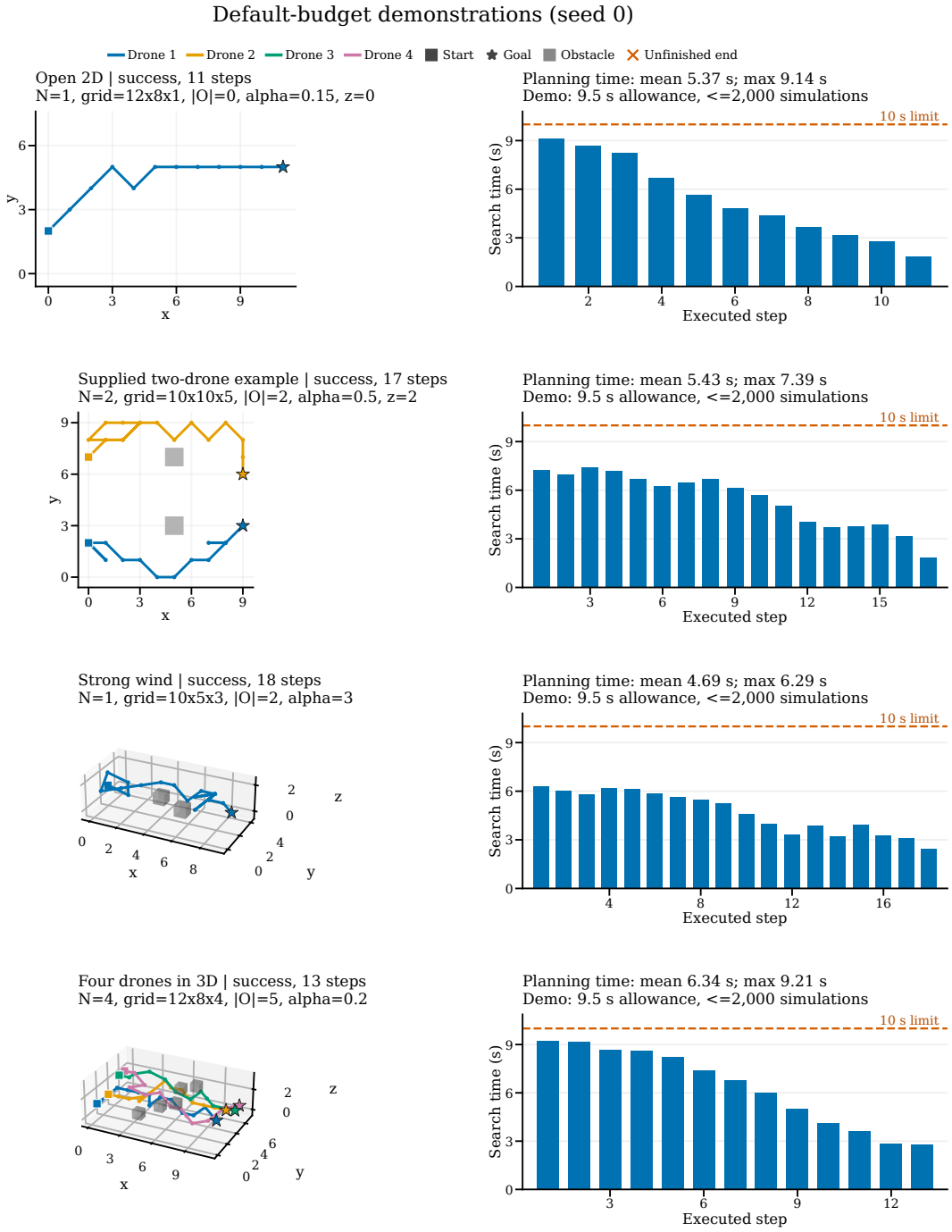


Figure 3: Default-budget demonstrations: open navigation, the supplied example, strong wind and four-drone 3D flight.

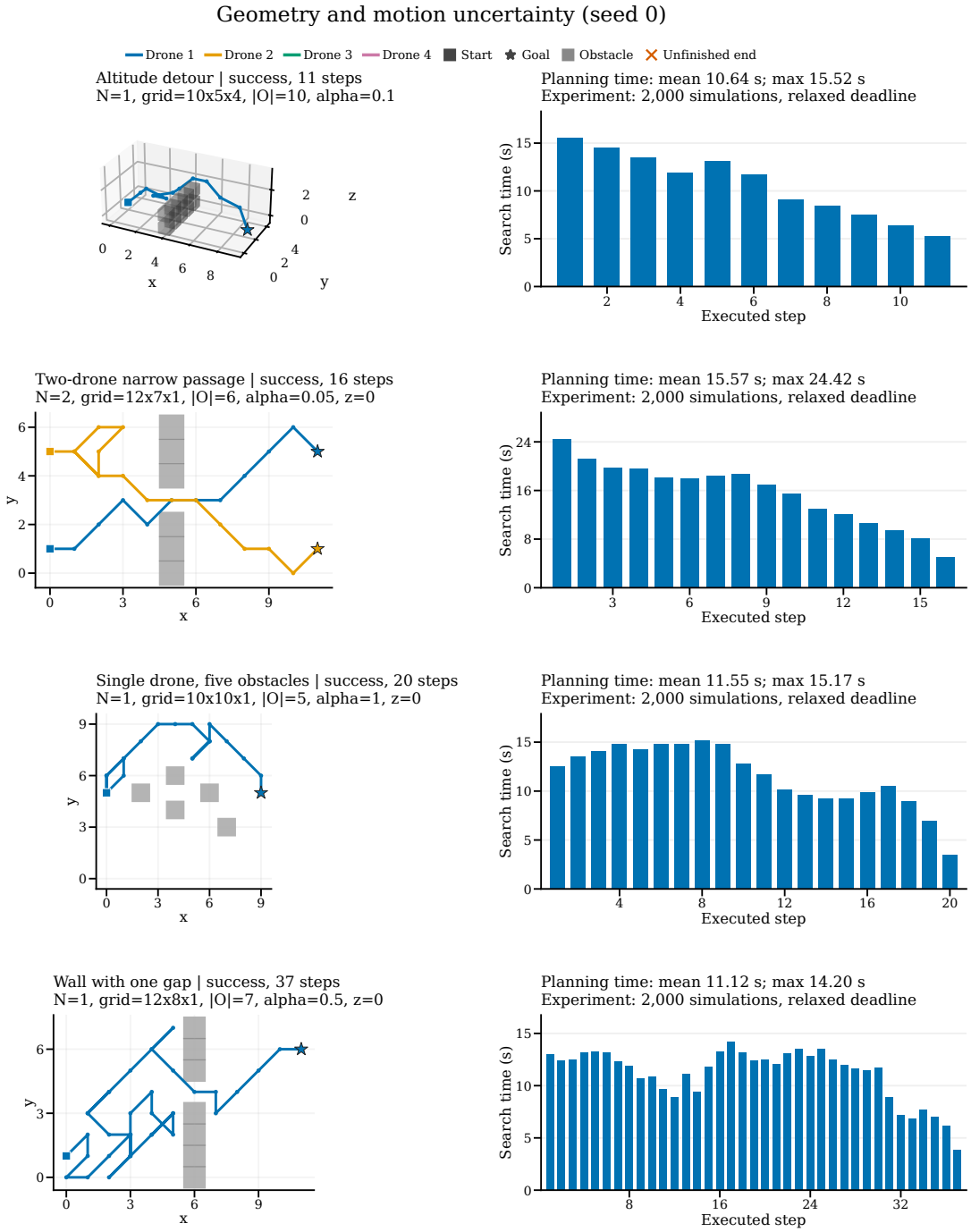


Figure 4: Fixed-simulation examples: altitude detours, narrow passage, clutter and a wall gap. The wall-gap layout has seven obstacles.

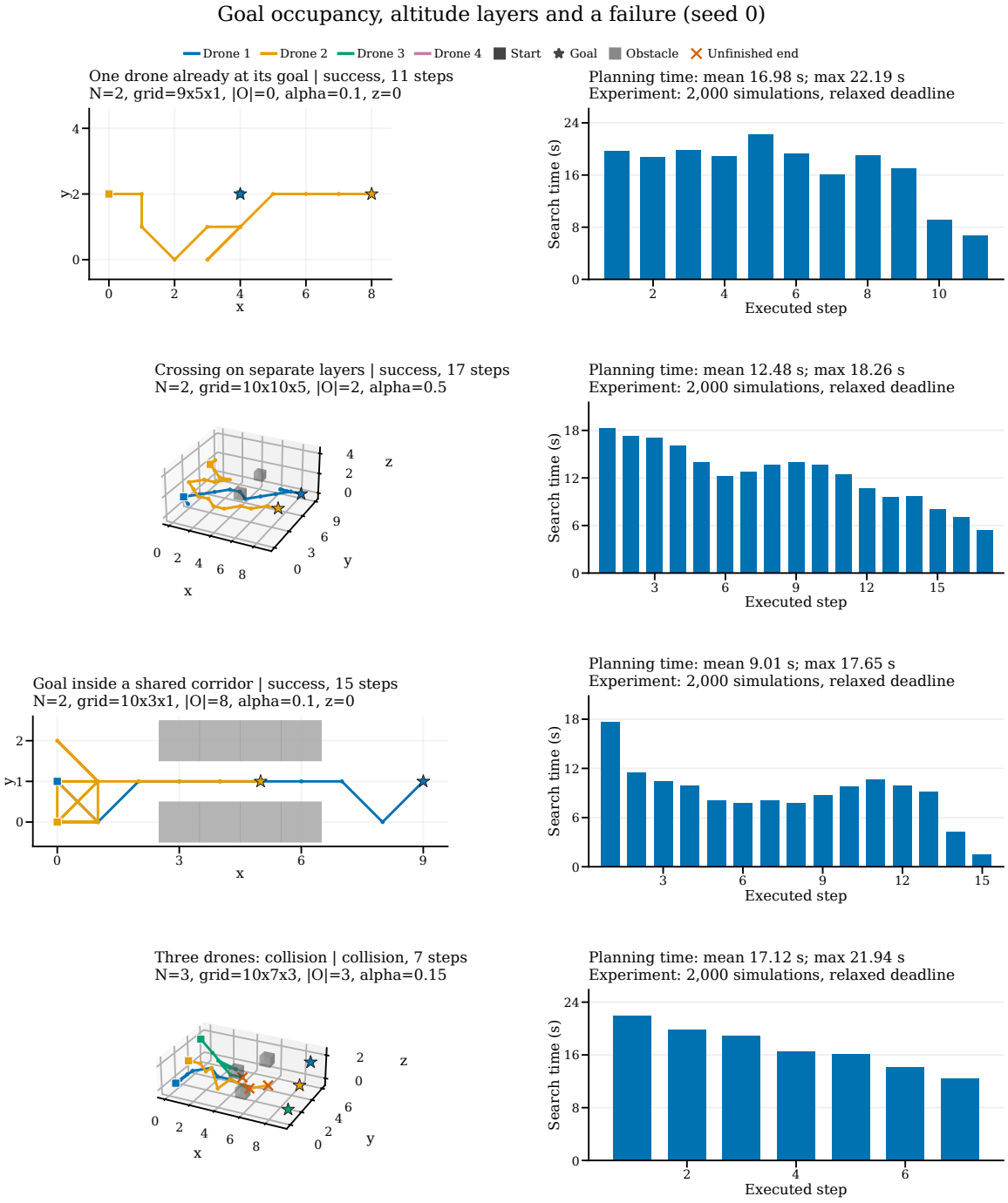


Figure 5: Fixed-simulation examples: an occupied goal, separate altitude layers, a shared corridor and a collision. The corridor has eight obstacles. Timings on this and the preceding sheet are not demo deadline measurements.